

Neutrino Mixing and Projective Particle Identity

A Structural Note

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Abstract

Neutrino oscillations establish that the weak flavour basis and the mass propagation basis of the lepton sector are not aligned. This note examines what this misalignment means for the ontology of particle identity within the Cosmochrony framework, where particles are invariants of a non-injective projection rather than primitive objects. Three structural observations are developed. First, basis misalignment is generic: neutral meson systems exhibit the same distinction between interaction labels and propagating eigenmodes, and the neutrino is distinguished only by being elementary within the Standard Model inventory. Second, within the projective reading, alignment — not misalignment — is the feature requiring a mechanism: an identity-stabilising projection channel. Charged fermions possess such channels; neutrinos, singlets under $SU(3)_C \times U(1)_{em}$, do not. The large PMNS mixing is then the unfixed default, consistent with the anarchy hypothesis, while the near-diagonality of the CKM matrix becomes the non-trivial explanandum, requiring the charged channels to pin the up-type and down-type mass bases to a common projective frame. Third, the Dirac-or-Majorana alternative is reread as the degree to which the projection resolves relational conjugation, with neutrinoless double beta decay as its standard discriminator. The note is structural throughout: it derives no mixing angles and no masses. It fixes the reading under which the neutrino sector is the sharpest available test case for projective particle identity, and states one falsifiable structural expectation: sectors with fewer identity-stabilising projection channels generically exhibit larger mixing between interaction and propagation bases. Interpretive status: the reinterpretation of particle identity as a projection invariant is a structural reading of standard neutrino phenomenology, not a competing prediction for it; all quantitative content of the standard oscillation formalism is retained as the operational effective description.

Keywords. neutrino oscillations; particle identity; projective invariants; PMNS matrix; CKM matrix; anarchy hypothesis; neutral meson mixing; Dirac and Majorana neutrinos; non-injective projection; ontology of particle physics.

Contents

1	Introduction	3
2	The Standard Misalignment	3
3	The Neutral-Meson Comparison	3
4	The Projective Reading: Alignment Requires a Stabiliser	4
5	The CKM–PMNS Contrast	4
6	Dirac versus Majorana as Conjugation Resolution	5
7	Phenomenological Outlook (Structural)	5
8	Positioning	5
9	Status of Claims	5
10	Conclusion	6

1 Introduction

The Standard Model tabulates particles as primitive entries: each species is a well-defined object carrying a fixed set of labels. Neutrinos are the elementary case where this tabulation is under visible strain. The states produced and detected by the weak interaction (ν_e, ν_μ, ν_τ) are not the states that propagate; the observed identity of a neutrino depends on which basis the observation accesses. This tension has recently been articulated as a pressure point on Standard-Model ontology in the philosophy of particle physics [12].

The Cosmochrony framework [2] treats particles as invariants of a non-injective projection Π from a static relational substrate, rather than as primitive objects. The white paper already assigns neutrinos a special position, as *partially projectable modes* close to the boundary of projectability. The present note develops the structural core of that position in the language of the fermionic-matter sub-programme [3], where the fermionic sector arises from the admissible spinor bundle $\mathcal{S}_\Pi$ and its projective endomorphism E_Π [4, 7].

The scope is deliberately narrow. The note does not derive mixing angles, masses, or oscillation corrections; the chiral polar factor carrying generation mixing is invisible to E_Π^2 at the present stratum [5], so any quantitative claim would be premature. What the note fixes is the reading: which features of the neutrino sector are generic, which require a mechanism, and which structural expectation is falsifiable.

2 The Standard Misalignment

The following facts are standard and are imported, not derived. Flavour eigenstates are defined by the charged-current weak interaction; mass eigenstates diagonalize propagation; the two bases are related by the unitary PMNS matrix U_{PMNS} ,

$$|\nu_\alpha\rangle = \sum_i (U_{\text{PMNS}}^*)_{\alpha i} |\nu_i\rangle, \quad \alpha \in \{e, \mu, \tau\},$$

a structure anticipated by Pontecorvo [16] and by Maki, Nakagawa and Sakata [15], and established experimentally by the observation of atmospheric and solar neutrino oscillations [9, 1]. Oscillations imply non-degenerate masses and therefore force an extension of the minimal Standard Model, in which neutrinos are massless. In the quark sector the analogous misalignment between up-type and down-type mass bases is encoded by the CKM matrix [8, 13], which is experimentally near-diagonal, whereas U_{PMNS} contains two large angles.

Operationally, the standard oscillation formalism — interference between mass eigenstates with distinct dispersion — is retained without modification throughout this note. The question addressed here is not *how* neutrinos oscillate but *what oscillation says about the ontology of the oscillating object*.

3 The Neutral-Meson Comparison

Basis misalignment is not exceptional in itself. Neutral meson systems exhibit the same structural distinction: K^0 and $\bar{K}^0$ are the strong-interaction (production) labels, while the propagating eigenmodes are their mixtures, as recognised by Gell-Mann and Pais [10]. Whenever an interaction basis and a propagation basis coexist and no mechanism forces them to coincide, the generic situation is that they do not.

The ontological difference is that neutral mesons are composite: their mixing can be attributed to the dynamics of constituents, and the naive particle ontology can retreat to the quark level. Neutrinos are elementary within the Standard Model inventory, so no such retreat is available. The neutrino is therefore the case where the contextuality of particle identity — flavour at production and detection, mass in propagation — must be confronted at the level of the fundamental tabulation itself.

4 The Projective Reading: Alignment Requires a Stabiliser

In the Cosmochrony framework a particle label is an invariant of the projection Π : flavour is an invariant of the weak coupling, and mass is a spectral invariant of the projected residue, the eigenvalue data of the projective endomorphism E_Π [7]. These are distinct invariants of distinct structures, and there is no structural reason for the weak flavour basis to diagonalize the projected spectral mass residue. The structural statement of this note is therefore:

Basis misalignment is the generic situation; basis alignment requires an additional projective stabiliser.

Neutrino oscillations are then not an anomaly of particle identity but the expected manifestation of a neutral chiral matching mode whose weak-interaction label and spectral mass label are distinct projective invariants.

This restates, in the vocabulary of the fermionic-matter sub-programme, the white-paper description of neutrinos as partially projectable modes [2]. One white-paper formulation is deliberately softened here. The claim that oscillations reflect “variations in projectability rather than interference between fundamental mass eigenstates” is retained only in its ontological part: the standard interference description remains the operational effective description, and what the framework reinterprets is the status of the mass eigenstates — projected spectral invariants rather than fundamental degrees of freedom.

5 The CKM–PMNS Contrast

The projective reading reverses the usual explanatory burden. As a structural reading, not a derived result, charged fermions possess projection channels that stabilise their identity between production and detection — the electromagnetic and colour couplings, together with the operational pinning provided by large mass splittings and rapid decoherence — whereas neutrinos, singlets under $SU(3)_C \times U(1)_{em}$, lack any such stabilising channel.

The large leptonic mixing encoded by U_{PMNS} is then the unfixed default. This is consistent with the anarchy hypothesis for neutrino masses [11], which showed that the oscillation data are well accounted for by a mass matrix with no preferred structure. The projective reading adds to anarchy what anarchy does not supply: a reason for its sector selectivity. Randomness is expected precisely where no identity-stabilising channel constrains the basis, and only there.

Conversely, the near-diagonality of U_{CKM} becomes the non-trivial feature to be explained. In this reading, CKM near-alignment requires the additional assumption that the charged projection channels pin the up-type and down-type mass bases to a *common* projective frame. This common-frame assumption is the load-bearing hypothesis of the reversal: two sectors stabilised independently could remain mutually rotated, so stabilisation alone does not imply alignment. Its status is conditional, and its derivation is an open problem of the fermionic-matter sub-programme [3].

The reversal yields one falsifiable structural expectation:

Sectors with fewer identity-stabilising projection channels should generically exhibit larger mixing between interaction and propagation bases, unless an independent alignment mechanism is present.

The known pattern — large PMNS angles in the unpinned neutral sector, small CKM angles in the doubly pinned quark sector — conforms to this expectation; a future extension of the inventory (for example a sterile or otherwise unpinned state) would test it further.

6 Dirac versus Majorana as Conjugation Resolution

The Majorana alternative [14] asks whether the neutrino is distinct from its antiparticle. In the projective reading this is not a fundamental binary choice but a question about the resolution of the projection: a Dirac neutrino corresponds to a configuration whose conjugate projected realisations are distinguished by Π , a Majorana neutrino to one whose partial projectability collapses the conjugate classes [2]. Lepton number conservation is then an emergent bookkeeping of regimes where conjugate configurations remain distinguishable, and its violation becomes admissible exactly where the distinction is erased.

The standard discriminator, neutrinoless double beta decay, acquires a structural interpretation: it probes whether the projection resolves relational conjugation in the neutral chiral sector. No rate prediction is made here; the observation or exclusion of the process constrains the resolution structure of Π , not a parameter of this note.

7 Phenomenological Outlook (Structural)

The following directions are recorded as structural expectations, without quantitative content at this stage. Long-baseline coherence: as a mode without identity-stabilising channels, the neutrino is the natural probe of decoherence effects sensitive to global properties of the projected geometry; departures from standard oscillation patterns at ultra-long baselines would be the corresponding signature. Neutrinoless double beta decay: the conjugation-resolution reading of Section 6. Cosmological backgrounds: as the modes closest to the projection boundary, relic neutrinos are the observational window most sensitive to pre-geometric structure [2]. Deriving any of these quantitatively requires the explicit Lorentzian E_Π and the Yukawa sector, which are the primary open deliverables of the fermionic-matter sub-programme [3, 6].

8 Positioning

Three adjacent positions delimit the present one. Hobart’s ontic-types analysis [12] identifies the same pressure point — neutrino mixing challenges a one-to-one correspondence between particle type, flavour label, and mass eigenstate — and proposes a family ontology in which the manifest state depends on interaction conditions. The present note shares the contextuality diagnosis but supplies a mechanism candidate: identity is a projection invariant, and its contextuality is the absence of a stabilising channel, embedded in a framework where the fermionic sector itself is derived [4]. The anarchy hypothesis [11] shares the “large mixing is the default” conclusion; the present reading adds the sector-selectivity criterion and the reversal of the explanatory burden onto CKM. Standard quantum field theory treats flavour and mass bases as effective bookkeeping related by unitary rotation; the present note retains that formalism operationally and differs only in the ontological status assigned to the bases.

9 Status of Claims

Standard (imported): flavour–mass basis misalignment, oscillations, neutral meson mixing, near-diagonal CKM, large-angle PMNS, anarchy consistency of the oscillation data. Structural (this note): the stabiliser reading of alignment, the identification of the neutrino as a neutral chiral matching mode, the PMNS-default/CKM-explanandum reversal, the conjugation-resolution reading of the Dirac–Majorana alternative, and the mixing-versus-channels expectation. Conditional: the common projective frame assumption for CKM near-alignment (load-bearing, underived). Open: the quantitative pinning mechanism, the derivation of mixing magnitudes (blocked at the trivial chiral polar class $[U_\Pi] = [I]$ of the present stratum [5]), the resolution criterion for the Dirac–Majorana alternative, and any oscillation correction.

10 Conclusion

Neutrinos are the elementary case where particle identity is manifestly contextual: the weak basis, the mass basis, and the ontological inventory do not coincide by necessity. Read projectively, this contextuality is not a puzzle but the default: identity labels are invariants of a non-injective projection, distinct invariants need not align, and alignment is what requires a mechanism. The neutrino sector thereby becomes the sharpest available test case for projective particle identity, and the quark sector inherits the substantive question: why is CKM nearly diagonal?

Interpretive outlook

The following is an interpretation, a structural reading of established phenomenology, and not a result. If particle identity is projective rather than primitive, the Standard Model table is a catalogue of stable projection invariants, and the neutrino marks its frontier: the entry least pinned by projection channels, closest to the boundary where the projected description itself degrades. On this reading, oscillation experiments are not merely measuring parameters of an extended Standard Model; they are probing how much identity a mode can retain when almost nothing in the projection holds it fixed. The interpretive wager of the framework is that the pattern of mixing across sectors — anarchic where unpinned, aligned where pinned — is a visible trace of the projection structure from which particles emerge.

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